

University of Texas at Austin 2025 Invitational Machines C Exam



Directions:

- **This is a class set, DO NOT WRITE**
- Write all answers on the answer sheets; any marks elsewhere will not be scored. If additional space is needed on a question, **clearly mark the question number and your team number on the top left of the page** and attach to your test.
- There is no penalty for wrong answers. Answer as many questions as possible, even if you aren't sure if you're correct. For questions with parts that build on one another, points will be awarded for correct answers on parts regardless of previous part completion.
- Units are required whenever giving a numerical answer. Don't worry about significant figures.
- Tiebreaker questions (in order):
 1. Section 4, Question 3
 2. Section 3, Question 19
 3. Section 3, Question 17
- Good luck on the test!

Section 1

Circle True or False; Each question is worth 2 points

1. True False In an ideal machine, input work always equals output work
2. True False The mechanical advantage of a wedge can be increased by making the wedge longer and thinner
3. True False A wheel and axle always provides a mechanical advantage greater than 1
4. True False Friction always increases the efficiency of a machine
5. True False A lever with the effort between the load and the fulcrum is a class 3 lever
6. True False A screw is a modification of an inclined plane
7. True False Work is defined as the product of force and displacement in the same direction
8. True False The coefficient of static friction is always less than the coefficient of kinetic friction
9. True False Actual Mechanical Advantage (AMA) is always less than or equal to Ideal Mechanical Advantage (IMA)
10. True False A simple machine can multiply both force and energy at the same time
11. True False The efficiency of a machine can be expressed as $\eta = \frac{AMA}{IMA}$
12. True False If a lever is balanced, the torques on both sides of the fulcrum must be unequal
13. True False For a pulley system, the IMA can be determined by counting the number of rope segments supporting the load
14. True False Potential energy is lost when an object is lifted vertically
15. True False The conservation of energy applies to all machines, regardless of efficiency
16. True False A block-and-tackle (compound pulley system) increases the distance the rope must be pulled compared to a single pulley
17. True False In a wheelbarrow, both a lever and a wheel-and-axle act together
18. True False The AMA of a compound machine is always exactly the product of the AMAs of its components
19. True False Combining two simple machines always guarantees higher efficiency
20. True False For a compound machine, the overall IMA is always less than the IMA of any single component machine
21. True False If a pulley system is self-locking, it means the friction in the pulleys is sufficient to prevent reverse motion under load

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|-----|------|-------|--|
| 22. | True | False | In a gear train, if the driver has fewer teeth than the driven gear, the driven gear always rotates faster |
| 23. | True | False | The angle of repose is numerically equal to the angle of friction |
| 24. | True | False | AMA can exceed IMA if friction in the machine assists motion |

Section 2

Circle the correct answer

1. (2 points) Which of the following is/are considered simple machine(s)?
 - A. Pulley
 - B. Wheel and axle
 - C. Screw
 - D. All of the above
2. (2 points) A wheelbarrow is an example of which class of lever?
 - A. Class 1
 - B. Class 2
 - C. Class 3
 - D. None of the above
3. (2 points) The IMA of an inclined plane is:
 - A. Height/Length
 - B. Length/Height
 - C. Thickness/Length
 - D. Force/Distance
4. (2 points) A knife functions primarily as which type of machine?
 - A. Lever
 - B. Pulley
 - C. Wedge
 - D. Screw
5. (2 points) A pulley system with 2 movable pulleys has an ideal mechanical advantage of:
 - A. 1
 - B. 2
 - C. 3
 - D. 4

6. (2 points) A screw with a pitch of 2 mm and a handle radius of 2.5 cm has how many turns per cm of linear advancement?
- A. 2
 - B. 4
 - C. 5
 - D. 10
7. (2 points) A ramp has $\text{IMA} = 5$. Its angle θ is closest to:
- A. 5.7°
 - B. 11.5°
 - C. 22.0°
 - D. 45.0°
8. (2 points) A pulley has mass and radius R . What happens if pulley inertia is included?
- A. Efficiency increases because pulleys add energy
 - B. Efficiency stays the same
 - C. Efficiency decreases due to rotational kinetic energy
 - D. AMA increases beyond IMA
9. (2 points) A wedge has separation angle 20° . Find its IMA.
- A. 1.37
 - B. 2.75
 - C. 5.5
 - D. 10
10. (2 points) A class 1 lever has an effort arm of 80 cm and a load arm of 20 cm. If the load is 100 N, what effort is required?
- A. 25 N
 - B. 50 N
 - C. 100 N
 - D. 400 N
11. (2 points) A block is pushed up a 4 m incline that rises 1 m. If the load is 200 N, what is the IMA of the incline?
- A. 0.25
 - B. 1
 - C. 4
 - D. 200

12. (2 points) A pulley system lifts a 300 N load with an effort of 75 N. If the IMA is 5, the efficiency is:
- A. 60%
 - B. 70%
 - C. 80%
 - D. 100%
13. (2 points) A gear train has driver gear with 20 teeth and driven gear with 60 teeth. If the driver rotates at 180 rpm, the driven rotates at:
- A. 180 rpm
 - B. 120 rpm
 - C. 90 rpm
 - D. 60 rpm
14. (2 points) A block of mass 10 kg is lifted 2 m with a pulley requiring 250 J of input work. What is the efficiency? ($g = 9.8$)
- A. 65%
 - B. 78%
 - C. 85%
 - D. 100%
15. (2 points) In an ideal machine:
- A. Input force = Output force
 - B. Input distance = Output distance
 - C. Input work = Output work
 - D. Input power = Output power
16. (2 points) For a lever in equilibrium:
- A. Sum of forces = 0
 - B. $IMA = AMA$
 - C. Sum of torques = 0
 - D. Output work > Input work
17. (2 points) AMA is defined as:
- A. Input distance / Output distance
 - B. $IMA \times \text{efficiency}$
 - C. Output force / Input force
 - D. Input work / Output work

18. (2 points) The coefficient of friction is defined as:
- A. Input/Output work
 - B. Load/Effort
 - C. Frictional force / Normal force
 - D. Torque/Force
19. (2 points) A third-class lever always has:
- A. Effort between load and fulcrum
 - B. $IMA < 1$
 - C. Efficiency greater than 100%
 - D. Torque output greater than input
20. (2 points) The IMA of a screw depends on:
- A. Pitch only
 - B. Radius only
 - C. Circumference / Pitch
 - D. Mass of screw
21. (2 points) A 5 kg block slides down a frictionless incline of 30° from a height of 2 m. Its velocity at bottom is closest to:
- A. 2.0 m/s
 - B. 6.0 m/s
 - C. 6.3 m/s
 - D. 9.8 m/s
22. (2 points) A lever has an AMA less than its IMA. The most likely cause is:
- A. Incorrect fulcrum position
 - B. Frictional forces and energy losses
 - C. Torque imbalance
 - D. Excess efficiency
23. (2 points) In a rope-and-pulley system with $IMA = 6$, the load moves:
- A. The same distance as the effort
 - B. Half the distance of the effort
 - C. One-sixth the distance of the effort
 - D. Six times the distance of the effort

24. (2 points) A wedge with length L and width t has IMA:
- A. t/L
 - B. L/t
 - C. $L \times t$
 - D. $t - L$
25. (2 points) A block on an incline begins to slide when angle = 25° . The coefficient of static friction is approximately:
- A. 0.25
 - B. 0.47
 - C. 0.66
 - D. 0.93
26. (3 points) A wheelbarrow is an example of which compound machine?
- A. Lever and inclined plane
 - B. Lever and pulley
 - C. Lever and wheel-and-axle
 - D. Screw and wedge
27. (3 points) A system has an inclined plane of IMA = 4 and a pulley system of IMA = 3 used in series. The ideal combined IMA is:
- A. 7
 - B. $7/12$
 - C. 12
 - D. $1/12$
28. (3 points) A screw jack uses a lever handle of radius 20 cm to turn a screw with pitch 5 mm. What type of compound machine is this?
- A. Pulley–wedge combination
 - B. Wheel-and-axle with a screw
 - C. Lever with wedge
 - D. Screw with inclined plane

29. (3 points) A gear train has driver gear A (30 teeth) driving gear B (90 teeth), which drives gear C (15 teeth). If A rotates at 12 rad/s, the speed of C is:
- A. 3 rad/s
 - B. 4 rad/s
 - C. 9 rad/s
 - D. 18 rad/s
30. (3 points) A screw jack with pitch $p = 5.0$ mm and handle length 0.40 m lifts a 2000 N load. If efficiency is 25%, the required input force is:
- A. 20 N
 - B. 50 N
 - C. 160 N
 - D. 400 N
31. (4 points) A lever of $\text{IMA} = 4$ drives a pulley system of $\text{IMA} = 3$. However, the lever arm is angled 30° relative to the vertical force applied, and the pulley has an additional 20% frictional loss. What is the effective overall mechanical advantage of the compound machine?
- A. 7.2
 - B. 9.6
 - C. 10.4
 - D. 12.0
32. (4 points) A two-stage compound machine has Stage 1 efficiency $\eta_1 = 70\%$ and Stage 2 efficiency $\eta_2 = 80\%$. Stage 1 is a lever with $\text{IMA} = 4$, Stage 2 is a pulley with $\text{IMA} = 3$. If the input effort is 120 N, which load is closest to what the system can actually lift?
- A. 700 N
 - B. 800 N
 - C. 806 N
 - D. 1,152 N

33. (5 points) A compound machine has three sequential stages:

- Stage 1: Lever with $\text{IMA} = 3$, efficiency = 90%
- Stage 2: Inclined plane with $\theta = 15^\circ$, efficiency = 75%
- Stage 3: Pulley system with $\text{IMA} = 2.5$, efficiency = 80%

If the input force is 100 N, what is the approximate maximum output load the system can lift?

- A. 370 N
- B. 400 N
- C. 425 N
- D. 450 N

Section 3

Write out a short response to the following questions

1. (4 points) A lever has an effort arm of 80 cm and a load arm of 20 cm. If the load is 100 N, what effort force is required?
2. (4 points) A ramp is 4.0 m long and rises 1.0 m. If a 200 N load is pushed up, calculate:
 - (a) The IMA of the ramp
 - (b) The effort force required (ignore friction)
3. (4 points) A pulley system lifts a 300 N load with an effort force of 75 N.
 - (a) What is the AMA?
 - (b) If the IMA is 5, what is the efficiency?
4. (6 points) A wheel of radius 0.6 m is attached to an axle of radius 0.1 m. A 200 N force is applied to the wheel rim.
 - (a) What is the IMA?
 - (b) Find the AMA if 900 N load is lifted.
 - (c) Given the previous part, what is the efficiency?
5. (6 points) A class II lever of total length 3.0 m supports a 500 N load placed 1.0 m from the fulcrum.
 - (a) What is the IMA?
 - (b) If the effort is applied 2.5 m from fulcrum and 220 N is needed, find the AMA.
 - (c) Given the previous part, what is the efficiency?
6. (8 points) A 20 kg mass is connected to a 40 kg mass via a pulley system with $IMA = 3$.
 - (a) If released (frictionless tabletop under the 20 kg), which way does the system accelerate?
 - (b) Compute the acceleration magnitudes of the 20 kg and 40 kg masses.
 - (c) If coefficient of friction under the 20 kg mass is $\mu = 0.20$, recalculate the accelerations.
7. (8 points) A 15 kg block slides down a rough incline 5.0 m long, 30° angle, $\mu = 0.15$.
 - (a) Determine the work done by friction.
 - (b) Compute the final velocity at the bottom of the block (start from rest).
 - (c) If connected over a pulley to lift a 5.0 kg block vertically, does the system move? Justify.
8. (5 points) A single-start screw has pitch $p = 1.5$ mm and a handle radius $R = 3.0$ cm. Find the IMA and the effort to raise a 350 N load at 80% efficiency.

9. (6 points) A gear train has driver A (20 teeth) driving B (50 teeth) on the same shaft as C (10 teeth) driving D (100 teeth). If A turns at 6.0 rad/s, find:
- (a) the overall speed ratio ω_D/ω_A
 - (b) ω_D
 - (c) the overall IMA (force multiplication), assuming ideal gears
10. (5 points) A symmetric wedge of total separation angle 2α splits a material. Show that $\text{IMA} = \frac{1}{\tan \alpha}$ and compute the IMA for $2\alpha = 18^\circ$.
11. (6 points) A crate (600 N) is lifted using a ramp (length 5.0 m, height 1.0 m) and a pulley system ($\text{IMA} = 2$).
- (a) Find the IMA of the ramp.
 - (b) Find the combined IMA of the ramp + pulley system.
 - (c) Calculate the required effort force (ignore friction).
12. (6 points) A screw jack has handle radius 0.20 m and pitch 2.0 mm.
- (a) Find the IMA of the screw.
 - (b) If the handle acts as a lever arm, find the overall IMA.
 - (c) Determine the required input force to lift a 2000 N load at 70% efficiency.
13. (5 points) A gear train connects a wheel-and-axle ($\text{IMA} = 5$) to a pulley system ($\text{IMA} = 3$).
- (a) Compute the combined IMA.
 - (b) If a 1200 N load is lifted, calculate the ideal effort.
14. (6 points) A double-pulley system lifts a 500 N load with an input effort of 120 N.
- (a) Calculate the AMA.
 - (b) If the IMA of the system is 5, calculate efficiency.
 - (c) If the effort is increased to 150 N, determine the new acceleration of the load (mass = 50 kg).
15. (7 points) A wedge of angle 15° is used to lift a block of weight 800 N.
- (a) Derive the expression for IMA of the wedge.
 - (b) Compute the IMA numerically.
 - (c) If the actual input force required is 300 N, calculate AMA and efficiency.
 - (d) Explain why the wedge might be self-locking in this case.

16. (8 points) A gear train has driver gear A (20 teeth) driving B (40 teeth, same shaft as C), C (10 teeth) driving D (100 teeth).
- Derive the overall speed ratio ω_D/ω_A .
 - If A rotates at 20 rad/s, compute ω_D .
 - If the system is 80% efficient, compute the AMA if input force is 10 N and load is 150 N.
17. (6 points) A screw jack has a mean diameter of 5 cm and pitch of 1 cm.
- Derive the formula for IMA in terms of screw diameter and pitch.
 - Compute the IMA.
 - If a 1000 N load is lifted with an input effort of 60 N, calculate efficiency.
18. (10 points) A lever with geometric $\text{IMA}_\ell = 5$ transmits force to a pulley system with $\text{IMA}_p = 4$. The input force is applied at an angle ϕ relative to the direction *perpendicular* to the lever arm, so only the perpendicular component produces torque. The lever stage has efficiency $\eta_\ell = 85\%$, and the pulley stage has efficiency $\eta_p = 75\%$. Let the input force be $F_{in} = 50$ N.
- Derive the effective overall *geometric* IMA as a function of ϕ .
 - For $\phi = 30^\circ$, compute the *ideal* output load (ignore losses but include the angle effect).
 - With the given stage efficiencies, compute the *actual* output load for $\phi = 30^\circ$.
 - If the maximum input linear speed at the handle is $v_{in} = 0.20$ m/s, estimate the output speed v_{out} under the ideal geometric mapping (include the angle effect consistently).
19. (12 points) A compound machine consists of a wheel-and-axle driving a power screw. The wheel radius is $R_w = 0.50$ m, the axle/screw mean radius is $R_s = 0.10$ m, and the screw pitch is $p = 0.005$ m (single start). Assume the wheel is keyed to the screw (no gearing between axle and screw). The wheel-axle stage has efficiency $\eta_w = 90\%$ and the screw stage has efficiency $\eta_s = 40\%$ under load.
- Find the IMA of the wheel-axle stage.
 - Find the IMA of the screw stage.
 - Compute the total geometric IMA.
 - For an input force $F_{in} = 30$ N applied at the wheel rim, compute the *actual* output load.
 - Determine whether the screw is self-locking by evaluating the lead angle λ and comparing $\tan \lambda$ to a typical friction coefficient $\mu = 0.05$.

Section 4

Write out your answer to the following questions

1. (5 points) Explain why a compound machine may have a very high IMA but still be inefficient in practice.
2. (5 points) Describe how potential energy and kinetic energy trade off in a non-equilibrium pulley system when the load accelerates.
3. (5 points) Explain why in compound machines the total mechanical advantage multiplies, but the efficiency always decreases compared to each individual stage.
4. (6 points) Explain why a machine can increase force but cannot increase the total amount of work done. In your answer, include the relationship between input work, output work, and efficiency, and describe how this applies to the definition of Ideal Mechanical Advantage (IMA).
5. (7 points) Define both IMA and AMA. Derive the equation for efficiency in terms of these two quantities, and explain why AMA is always less than or equal to IMA.
6. (6 points) Derive the formula for the IMA of an inclined plane. State your assumptions clearly (ignore friction). Then explain how the efficiency would change if friction were included and how this affects the AMA.
7. (6 points) Explain how the Ideal Mechanical Advantage of a pulley system can be determined by counting rope segments. Why does this method work? Justify mathematically using the conservation of work principle.
8. (5 points) Why does friction always reduce the efficiency of a machine? *Hint: relate friction to input work, output work, and wasted energy.*
9. (6 points) Compare the effects of static and kinetic friction in simple machines. Which one determines whether the machine starts moving, and which one governs motion after it starts?
10. (6 points) Explain why increasing the force output of a machine always requires a proportional increase in the input distance. Derive this from the definition of work and show the force–distance tradeoff explicitly.